ISE LABS

HTOL Setup Information

For
Inphi

DEVICE #: Vega 23X23

DEVICE REV: BA
PACKAGE: 729 BGA 0.8mm
HTOL Equipment: HPB5

Last Updated: 09/25/2019
Prepared by: Harpreet

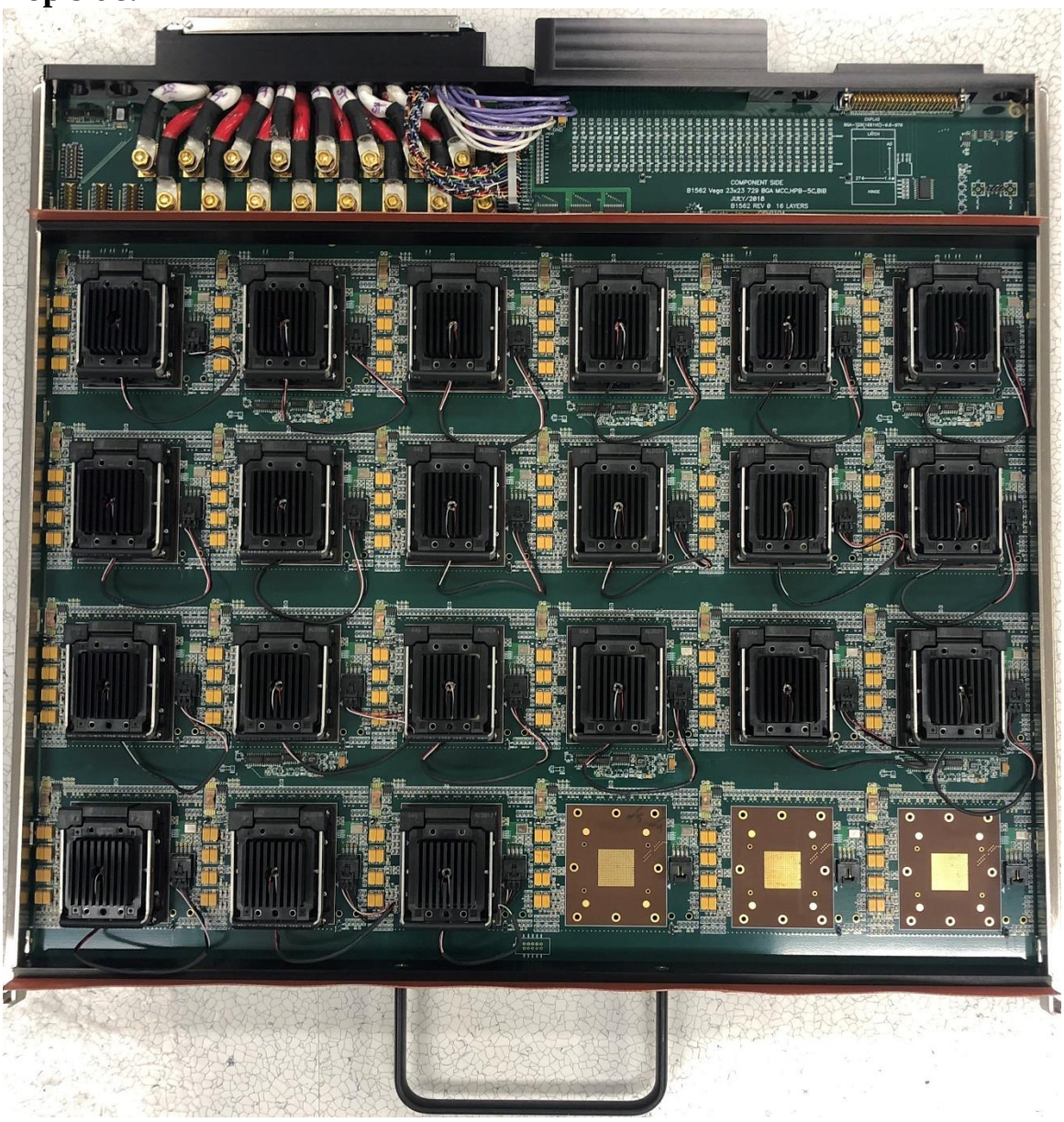
ISE LABS	**HTOL/ELFR** **setup information for** **Inphi Vega 23X23**

B/I Pattern information:

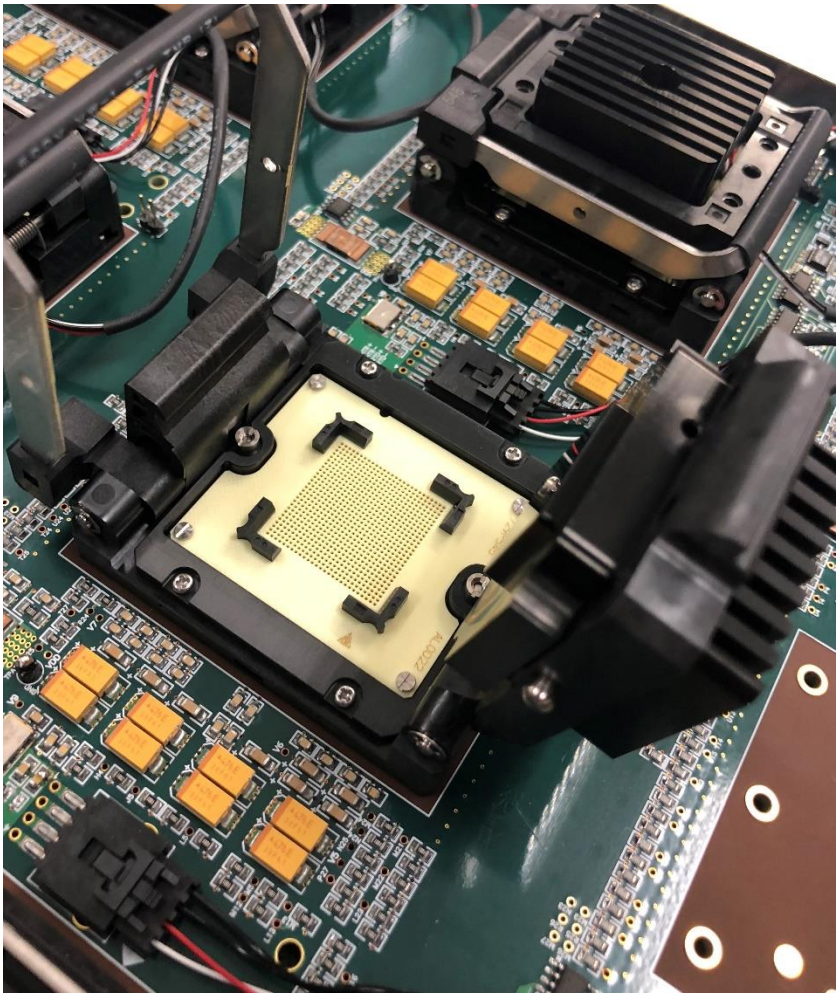
Pattern Name	*Program Name*
Clear.svm+fw_803_loopback.svm+mbist.svm+saf_edt_be__sdc_0627_0946_capture.svm	Inphi_B1562_Vega_23X23

Board Images

Top side:



DUT:



ISE LABS

HTOL/ELFR
setup information for
Inphi Vega 23X23

Device and Board Information

- 20 DUTs per board
- 80 DUTs total (4 boards)

HTOL Chamber Temperature:
Ta = 25C
Tc = 90C
Tj = 125C

Power Supply Information:

Supply	Name	Voltage	Seq. Up	Seq. Down	Delay
V0-V3	VDD (6 sockets)	0.88V	1	5	500ms or more
V4-V5	AVDD (6 sockets)	1.24V	2	4	500ms or more
V6	AVDD_PLL (20 sockets)	1.24V	3	3	500ms or more
V7	VDDIO (6 sockets)	1.24V	4	2	500ms or more
V8-V9	OSC (20 sockets)	2.2V	5	1	500ms or more

Current @ 25ºC (Tc):

Supply	Voltage	Current/Device (LB)	Current/Device (Mbist)	Current/Device (Scan)
V0-V3	0.88V	7.8A	0.2-0.4A	0.4A
V4-V5	1.24V	6.1A	0.1A	0.5A
V6	1.24V	3A	0.1A	0.1A
V7	1.24V	0.01A	0.01A	0.01A

Current @ 90ºC (Tc):

Supply	Voltage	Current/Device (LB)	Current/Device (Mbist)	Current/Device (Scan)
V0-V3	0.88V	8.3A	0.3-0.6A	0.7A
V4-V5	1.24V	6.5A	0.1A	0.9A
V6	1.24V	3.2A	0.1A	0.1A
V7	1.24V	0.01A	0.01A	0.01A

Full Board Noise Measurements:

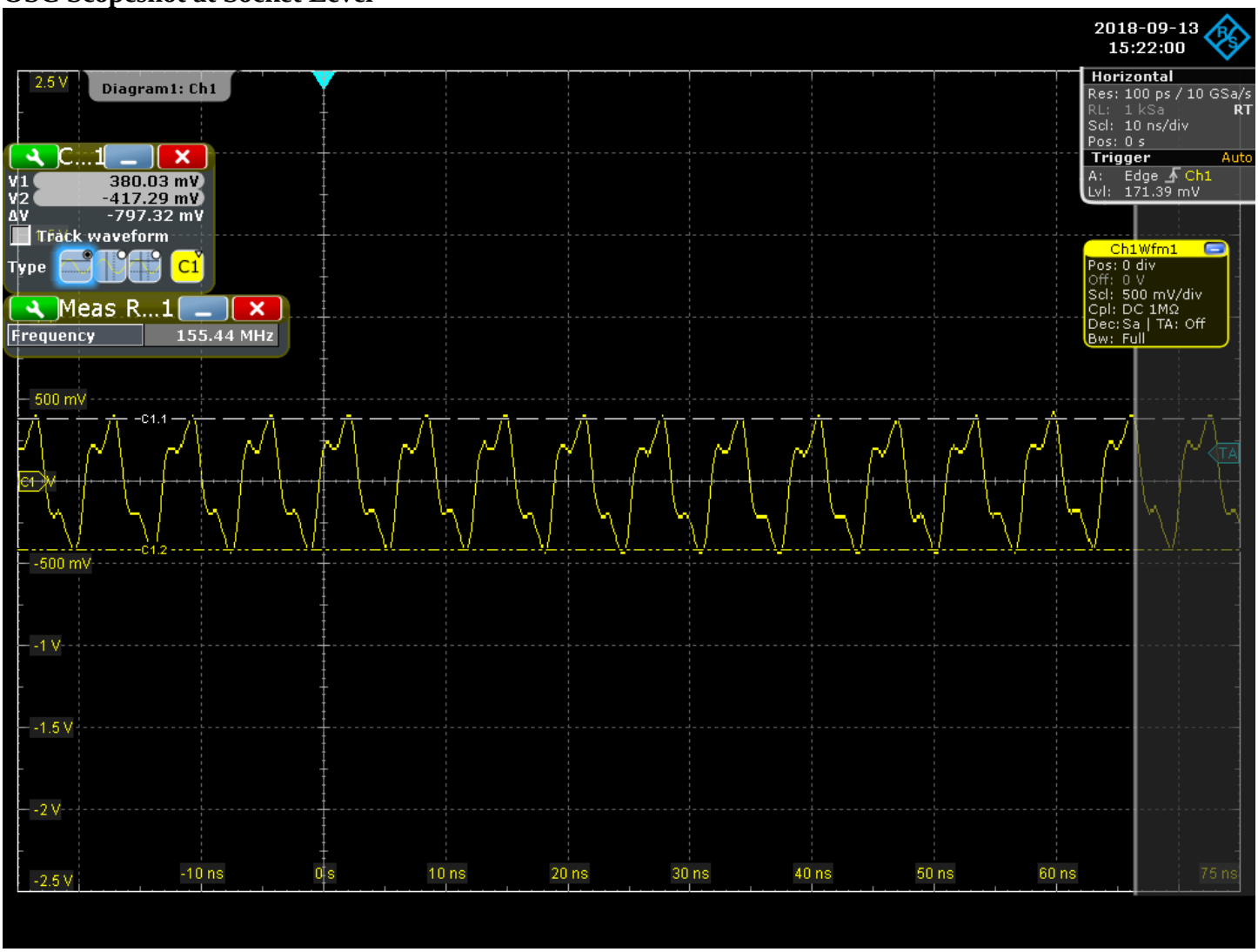
Supply	Min	Nominal	Max
V0-V3	0.86V	0.88V	0.89V
V4-V5	1.22V	1.24V	1.25V
V6	1.21V	1.24V	1.25V
V7	1.23V	1.24V	1.26V

Pattern Information

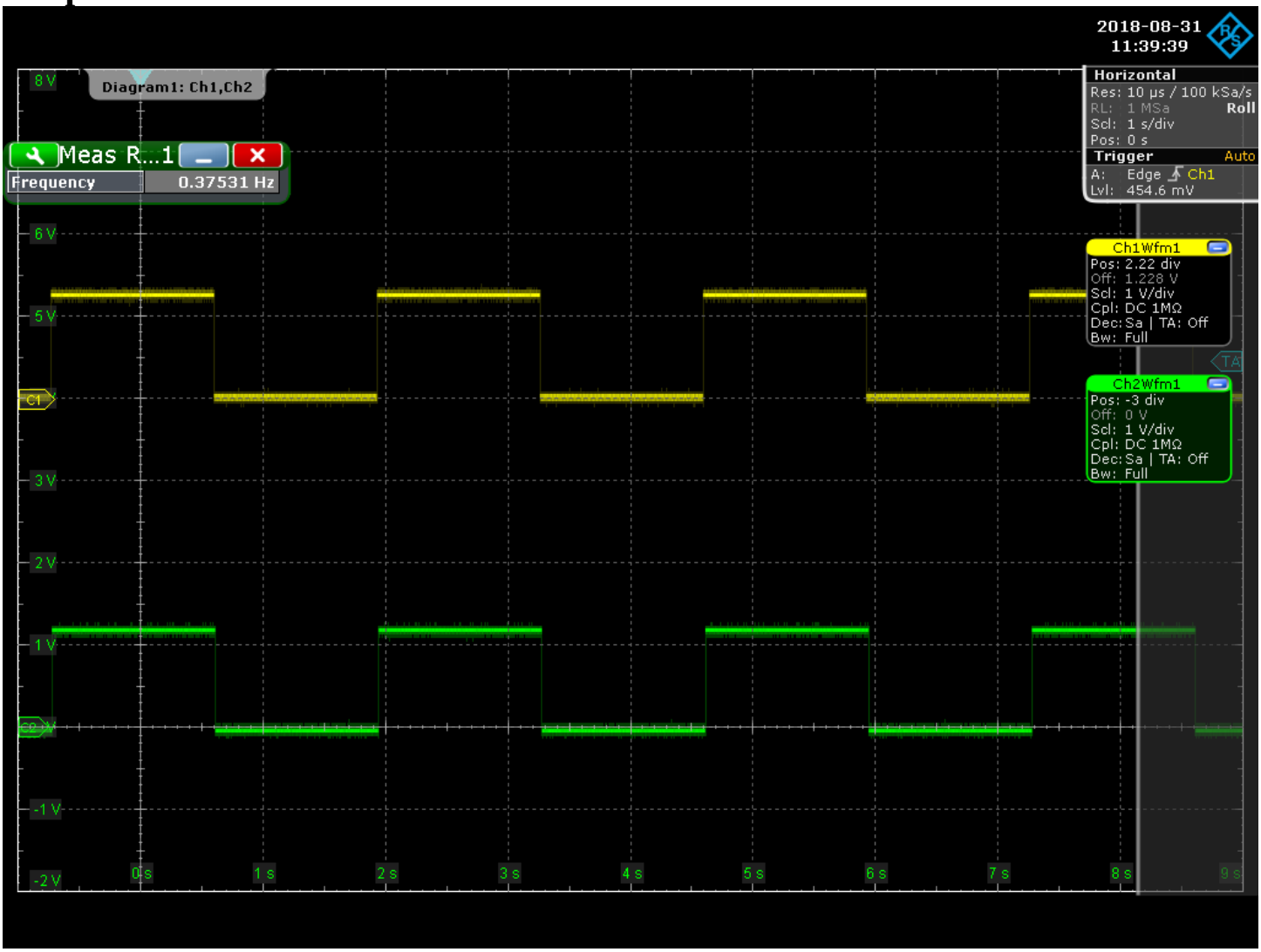
Pattern filenames and sequence:

Sequence	Pattern name	Hold Time
1	Clear	0 Minute
2	fw_803_loopback	1 Minute (on last vector)
3	mbist_P3_dig_top_slow	0 Minute
4	saf_edt_be__sdc_0627_0946_capture	0 Minute
5	Go back to # 1	

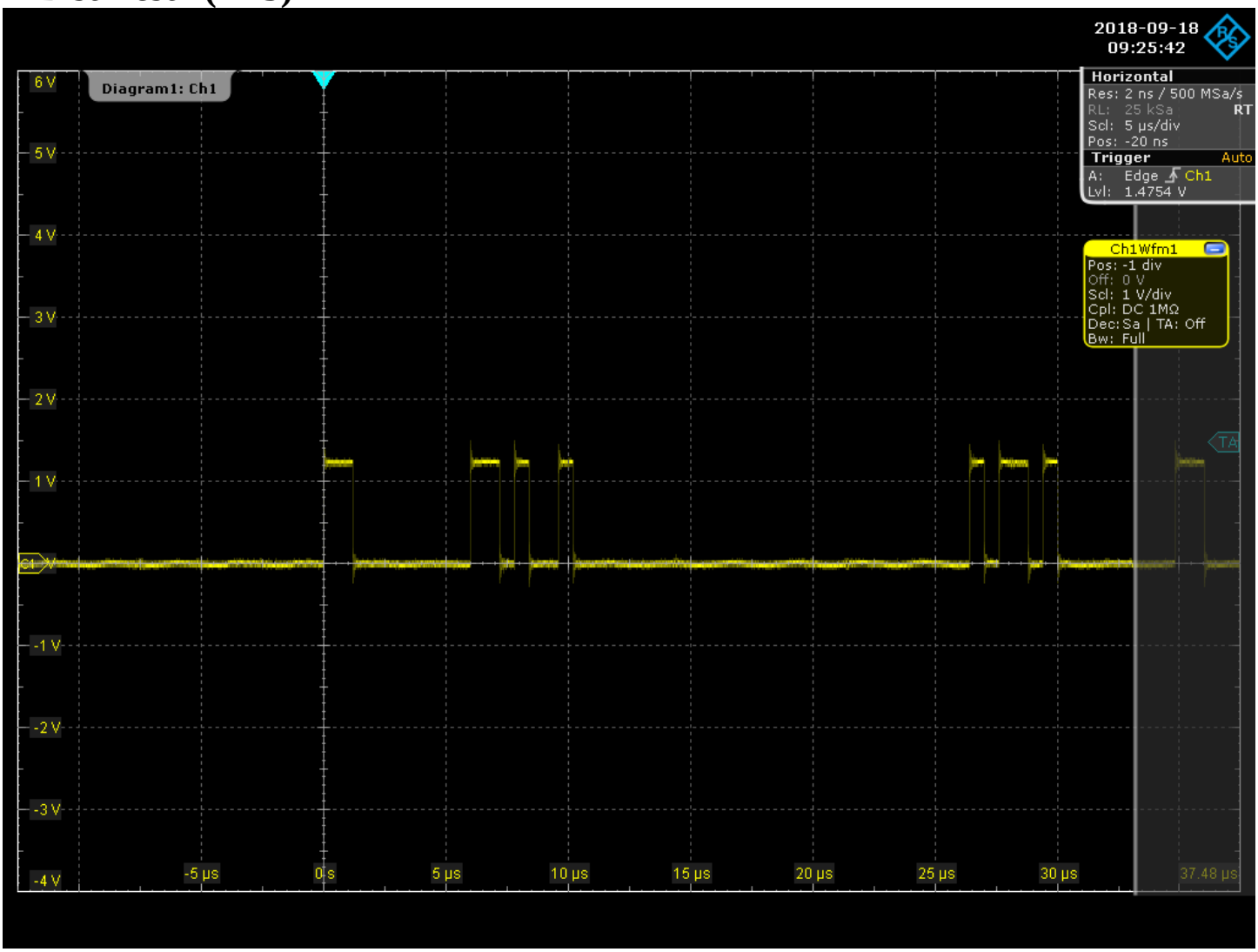
OSC Scopeshot at Socket Level



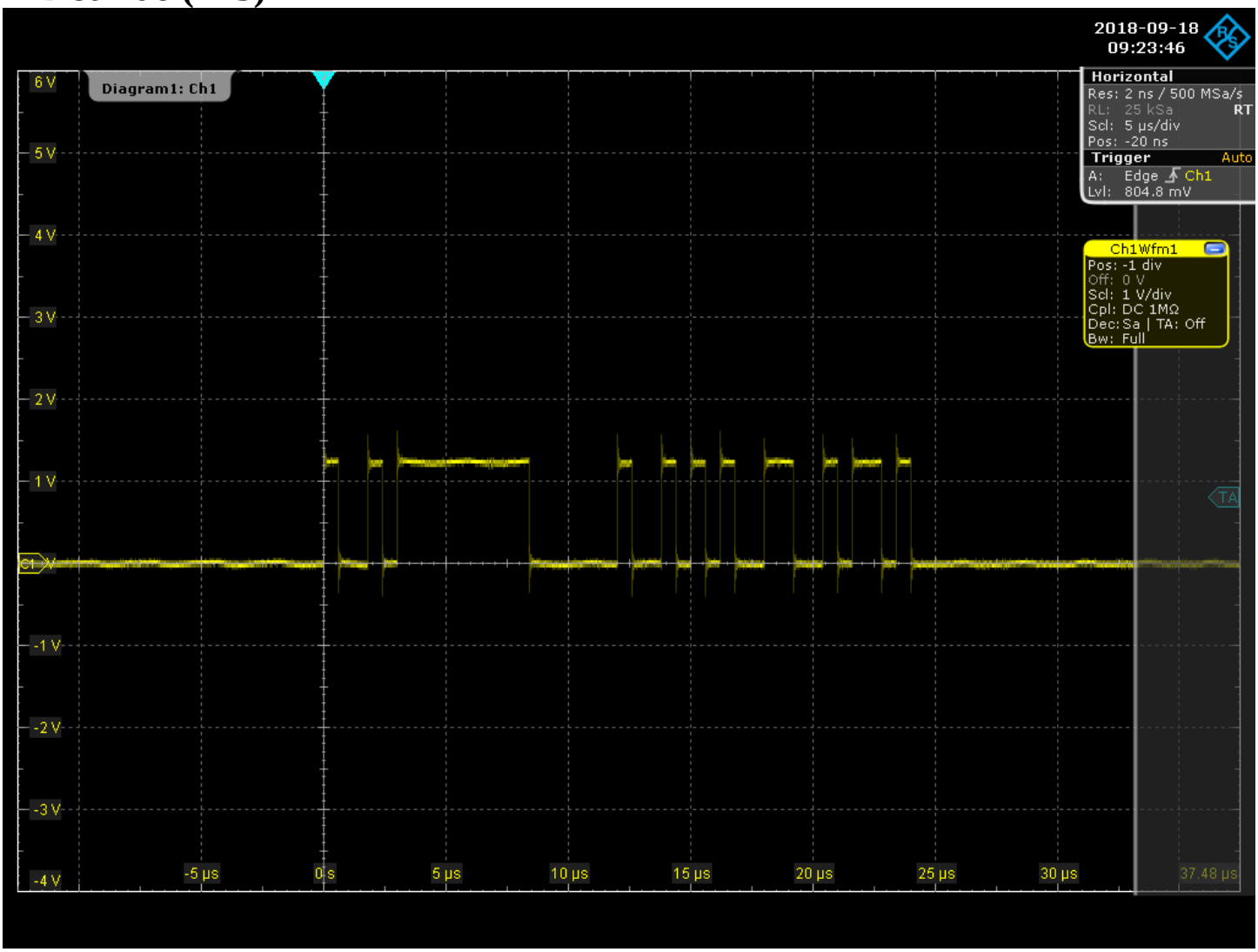
Outout Information
LoopBack GPIO1 & 3



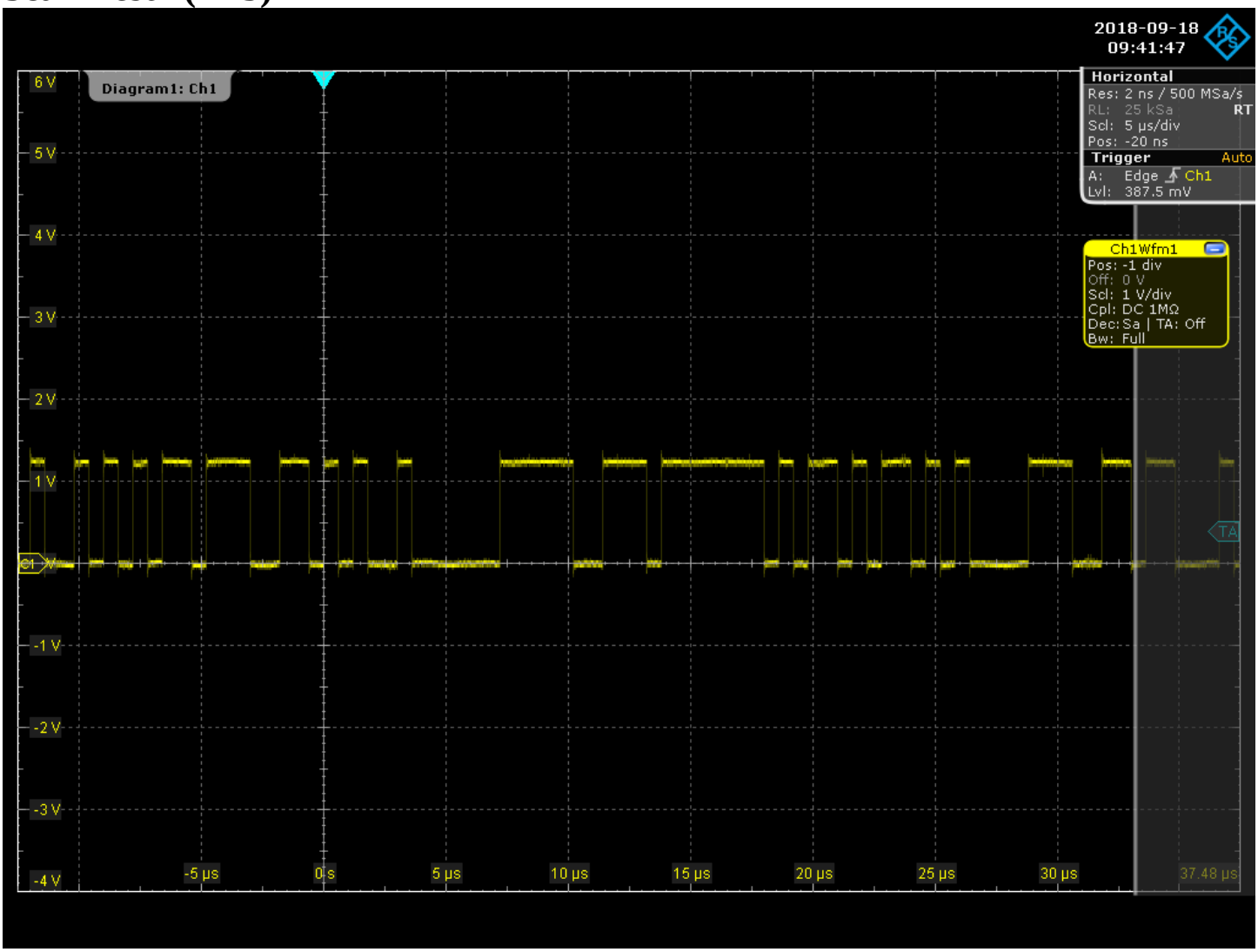
MBist Test1 (N25)



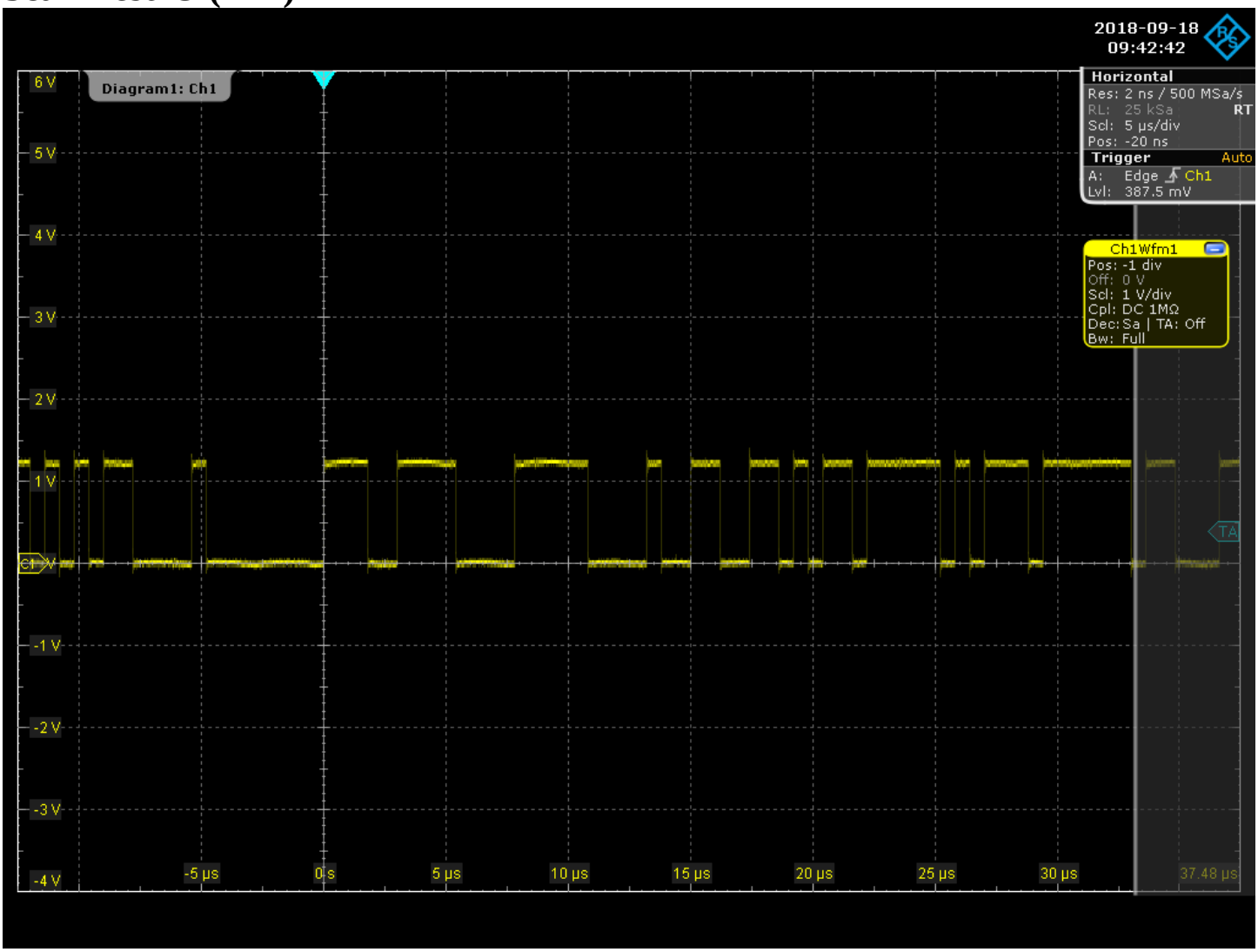
MBist Tdo (P25)



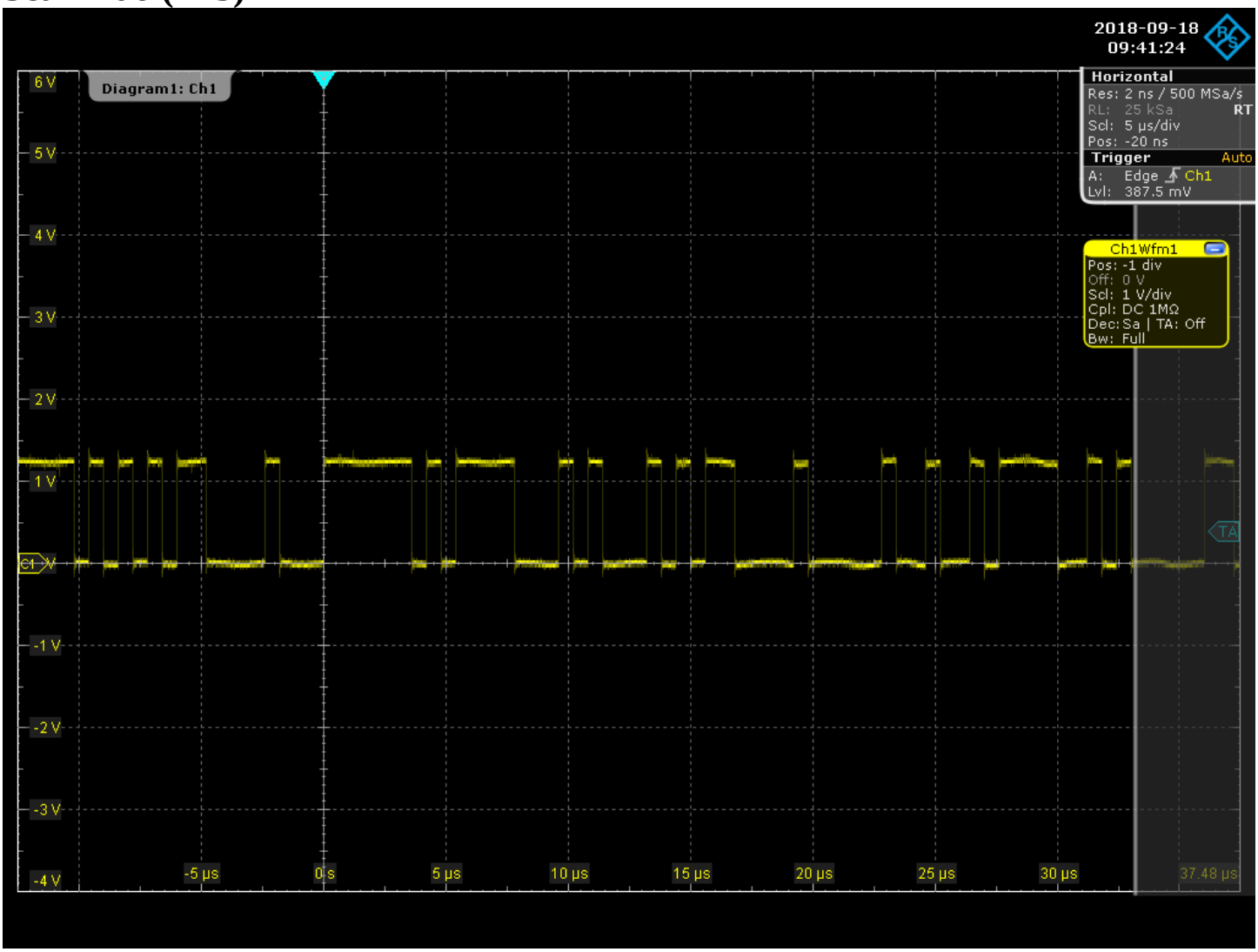
Scan Test1 (N25)



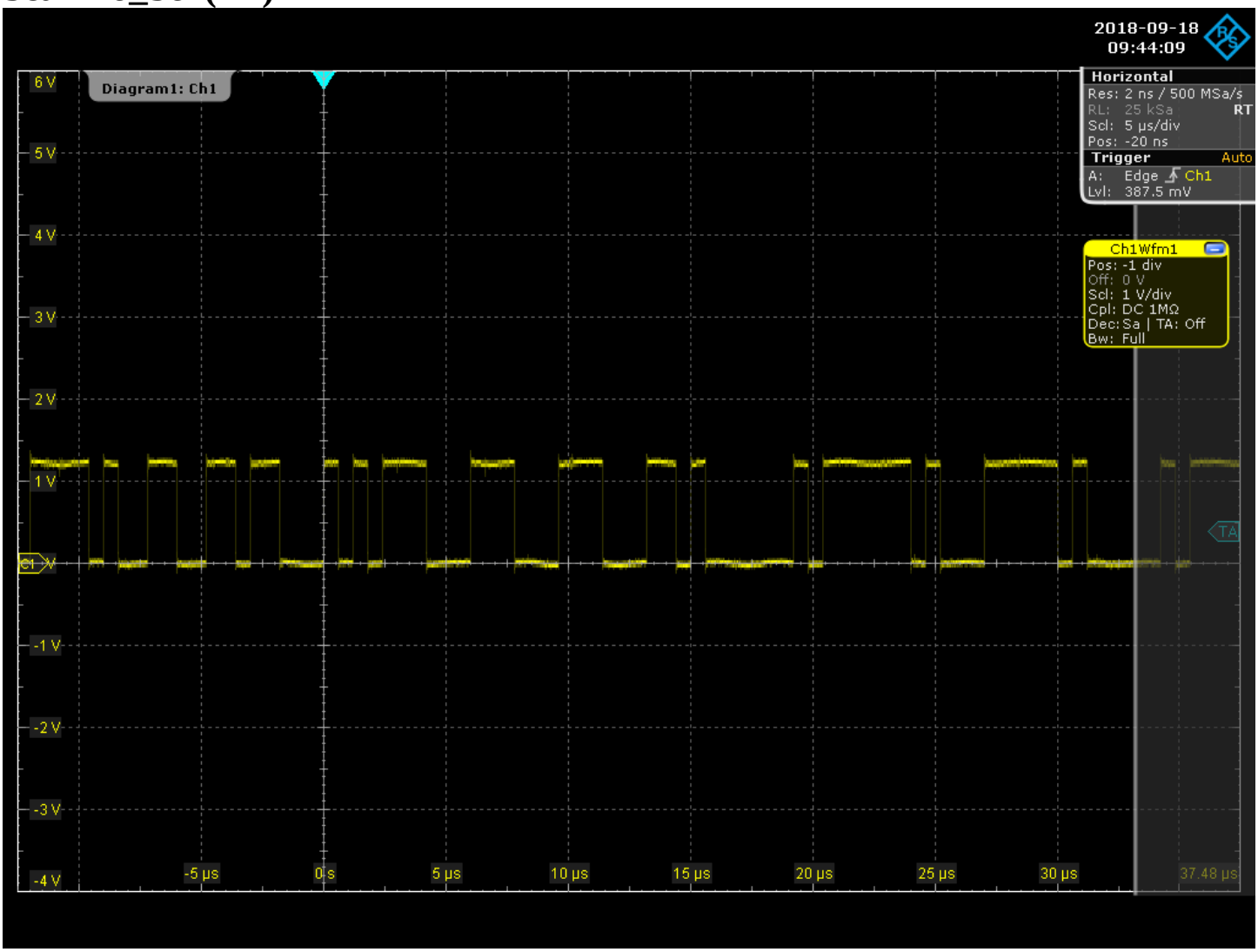
Scan TestIO (T12)



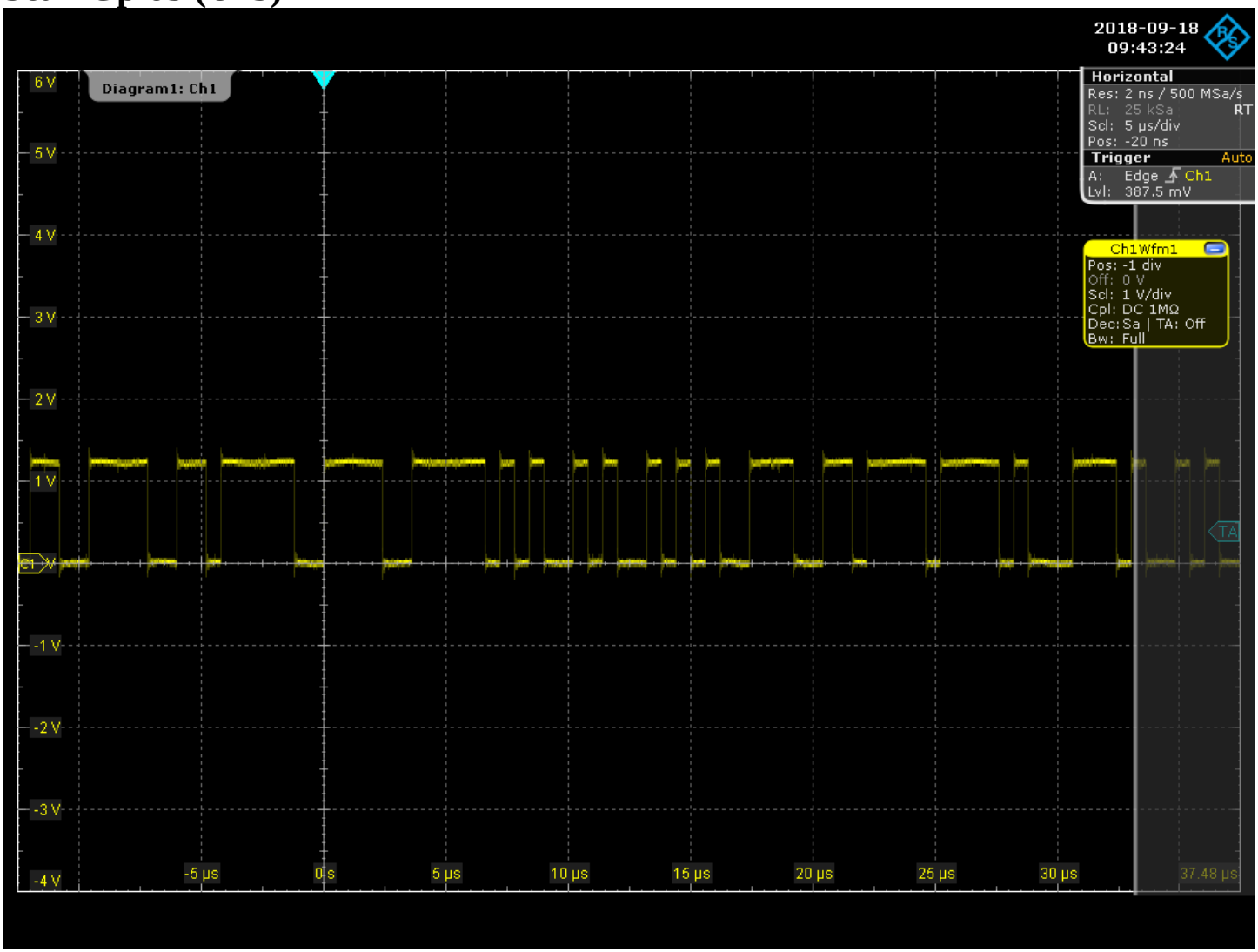
Scan Tdo (P25)


2018-09-18
09:41:24
Diagram1: Ch1
Horizontal
Res: 2 ns / 500 MSa/s
RL: 25 kSa
RT
Scl: 5 µs/div
Pos: -20 ns
Trigger
Auto
A: Edge Ch1
Lvl: 387.5 mV
Ch1Wfm1
Pos: -1 div
Off: 0 V
Scl: 1 V/div
Cpl: DC 1MΩ
Dec: Sa | TA: Off
Bw: Full
6 V
5 V
4 V
3 V
2 V
1 V
-1 V
-2 V
-3 V
-4 V
-5 µs
0 s
5 µs
10 µs
15 µs
20 µs
25 µs
30 µs
37.48 µs

Scan Po_Sel (R4)

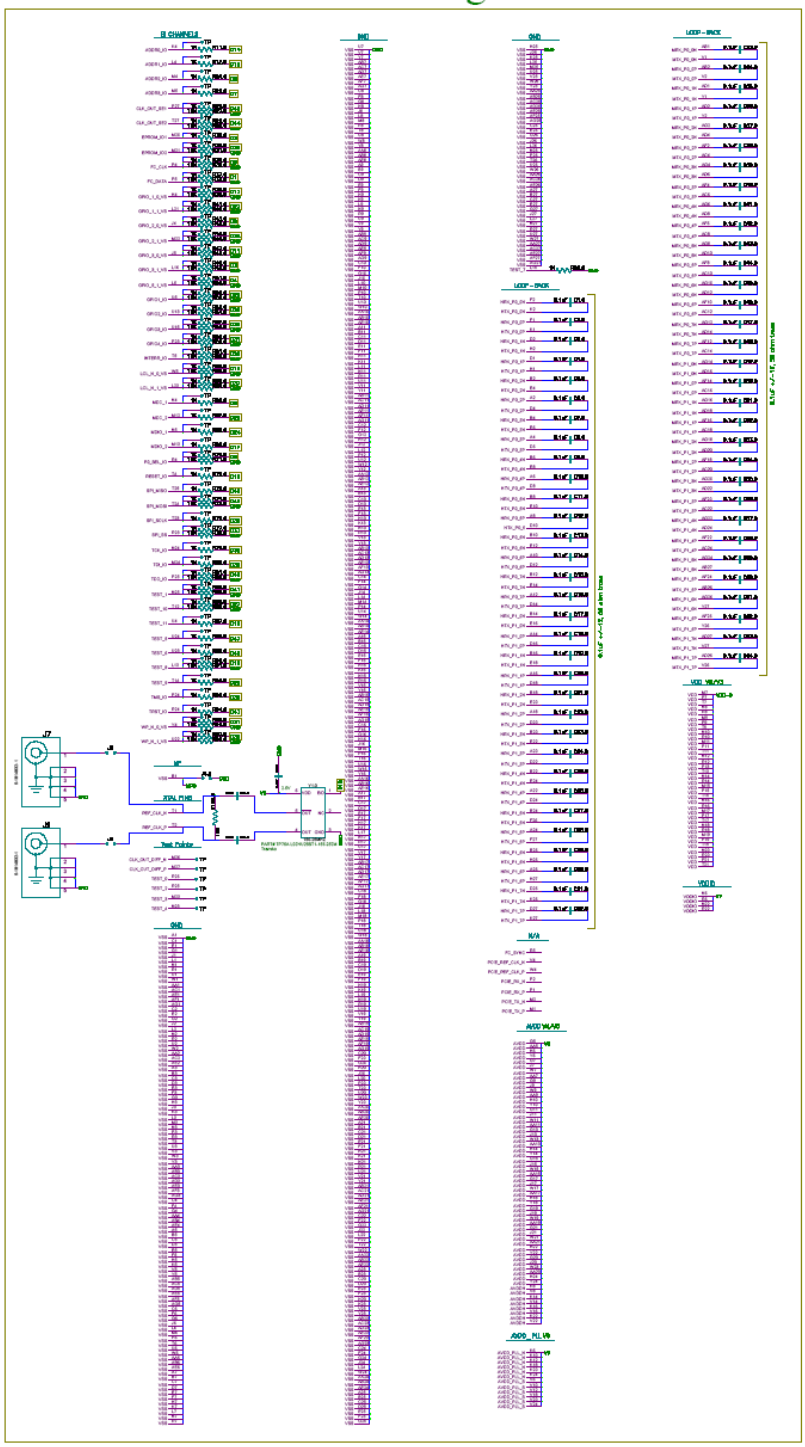


Scan Gpio3 (U15)



Cell Schematic

24 DUTS S0 B1562 Vega 23x23 729 BGA



Other Notes

Document Updates:

- Document created 09/19/2018
- Document updated 09/25/2018